1–13 The 12-V automobile battery in Figure P1–13 has an output capacity of 100 ampere-hours (Ah) when connected to a head lamp that absorbs 200 watts of power. Assume the battery voltage is constant.

(a) Find the current supplied by the battery.

(b) How long can the battery power the headlight?

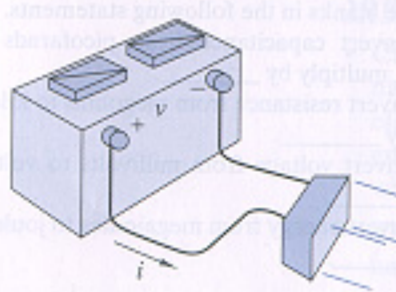

FIGURE P1–13

1–21 Two electrical devices are connected as shown in Figure P1–21. Using the reference marks shown in the figure, find the power transferred and state whether the power is transferred from A to B or B to A when

(a) $v = +33$ V and $i = -2.2$ A

(b) $v = -12$ V and $i = -1.2$ mA

(c) $v = +37.5$ V and $i = +40$ mA

(d) $v = -15$ V and $i = -43$ mA

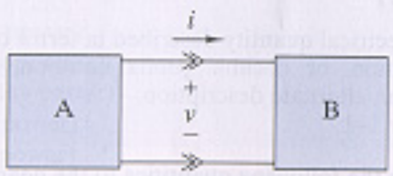

FIGURE P1–21

1–24 Using the passive sign convention, the voltage across a device is $v(t) = 5\cos(10t)$ V and the current through the device $i(t) = 0.5\sin(10t)$ A. Calculate the device power at $t = 0.2$ s and $t = 0.4$ s and state whether the device is absorbing or delivering power.

1–2 A 6.2-kΩ resistor dissipates 12 mW. Find the current through the resistor.

2–5 In Figure P2–5 find R_x and the power delivered to the resistor.

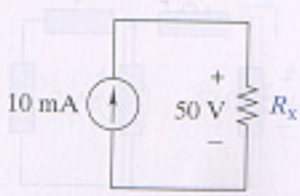

FIGURE P2–5

2–13 For the circuit in Figure P2–13:

(a) Identify the nodes and at least three loops in the circuit.

(b) Identify any elements connected in series or in parallel.

(c) Write KCL and KVL connection equations for the circuit.

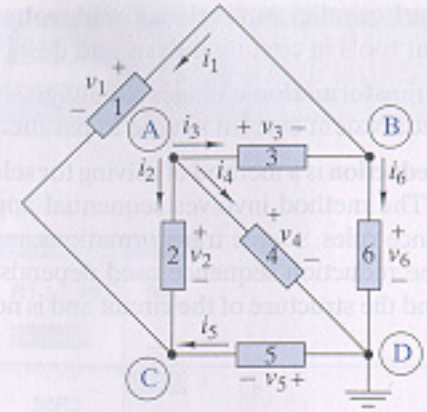

FIGURE P2–13

2–14 In Figure P2–13 $v_2 = 5$V, $v_3 = -8$V, and $v_4 = 3$V. Find v_1, v_5, and v_6.

2–20 Find v_x and i_x in Figure P2–20.

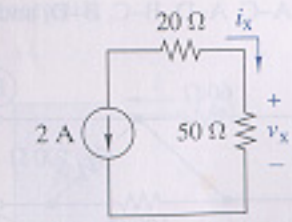

FIGURE P2–20

2–28 Find the equivalent resistance R_{EQ} in Figure P2–28.

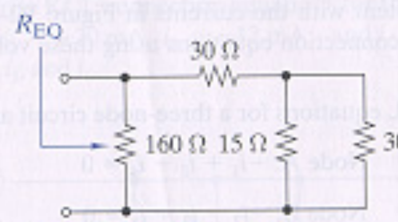

FIGURE P2–28

2–36 Select the value of R_x in Figure P2–36 so that $R_{EQ} = 60$ kΩ.

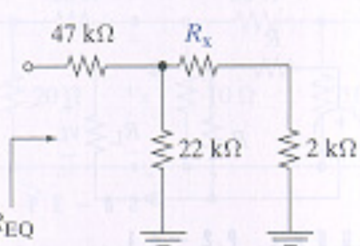

FIGURE P2–36